使用 Python

ONTAP Automation

NetApp
January 17, 2022

This PDF was generated from https://docs.netapp.com/zh-cn/ontap-automation/preparing_to_use_the_python_client_library.html on January 17, 2022. Always check docs.netapp.com for the latest.

目录

使用 Python 1
NetApp ONTAP Python 客户端库 1

使用 Python

Python 是一种受欢迎的编程语言，尤其适用于数据中心自动化。您可以使用 Python 的原生功能或通过 ONTAP Python 客户端库访问 NetApp REST API 。

NetApp ONTAP Python 客户端库

从 ONTAP 9.6 开始，您可以选择使用 Python 客户端库访问 ONTAP REST API 。该库可为客户端开发环境提供多种基础服务，例如连接管理，异步处理，异常处理和错误消息。下载并安装 ONTAP Python 客户端库软件包后，您可以快速创建强大的 Python 脚本，以支持 ONTAP 部署的自动化。

> NetApp 维护一个 GitHub 存储库，其中包含代码示例和其他有用信息。您可以导航到 *examples* 文件夹以使用 Python 客户端库访问示例。

"ONTAP REST Python GitHub 存储库"

正在准备使用 Python 客户端库

在使用 Python 客户端库之前，您应准备好环境。

基本要求

您必须使用 Python 3.5 或更高版本。此外，还需要以下软件包：

- 请求 2.22.1 或更高版本
- requests-toolbelt 0.9.1 或更高版本
- marshmallow 3.2.1 或更高版本

软件包名称和版本

Python 客户端库软件包的名称是 * 。 netapp-ontap* 。与软件包关联的版本是生成库时使用的 ONTAP 主要版本号和次要版本号的组合，以及 ONTAP 版本中客户端的次要版本。例如，有效版本号包括： 9.2.1 ， 9.6.2 和 9.1.1 。

安装

您必须使用 pip 从 Python Package Index （ Pypi ）网站安装 NetApp_ONTAP 软件包。

文档和其他资源

有关文档和其他资源的链接，请参见 _NetApp 开发人员网络 _ 。

"PyPI： NetApp ONTAP 9.10.1"

"适用于 9.10.1 的 NetApp PCL-文档"

"NetApp 开发人员网络： ONTAP RESTful API"

用于检索集群配置的脚本

以下脚本提供了一个简单的示例，用于说明如何使用 Python 客户端库。您可以在命令行界面中使用 Python 3 运行此脚本，以检索 ONTAP 集群配置。

```
##-----------------------------------------------------------------------
#
# Description: Python script to retrieve the cluster configuration.
#
# Usage example:
#
# python3 get_cluster.py
#
#
# (C) Copyright 2020 NetApp, Inc.
#
# This sample code is provided AS IS, with no support or warranties of
# any kind, including but not limited for warranties of merchantability
# or fitness of any kind, expressed or implied. Permission to use,
# reproduce, modify and create derivatives of the sample code is granted
# solely for the purpose of researching, designing, developing and
# testing a software application product for use with NetApp products,
# provided that the above copyright notice appears in all copies and
# that the software application product is distributed pursuant to terms
# no less restrictive than those set forth herein.
#
##-----------------------------------------------------------------------
# Global configuration for the library
from netapp_ontap import config
# Support for the connection to ONTAP
from netapp_ontap import HostConnection
# Specific API needed for this script
from netapp_ontap.resources import Cluster
# Create connection to the ONTAP management LIF
conn = HostConnection("10.236.252.97", username="admin",
password="mypassword", verify=False)
# Set connection as the default for all API calls
config.CONNECTION = conn
# Create new cluster object
clus = Cluster()
# Issue REST API call
clus.get()
# Display the cluster configuration
print(clus)
....
```

Copyright Information

Copyright © 2022 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system-without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.